Preregistration — Retrieval Section

Draft for review. Section numbering to be aligned with the parent document.

R1. Role of retrieval in the study design

Arm B represents a deployable guardrail: a retriever selects one safety checklist from the full corpus using the clinical vignette text alone, with no diagnosis, no case identifier, and no gold label supplied at any point. Arm C represents an oracle ceiling, in which the correct checklist is supplied directly by the researchers.

The two contrasts carry distinct meanings and are analysed separately:

- **Arm A** → **Arm B** estimates the value of a guardrail that could be deployed.
- **Arm B** → **Arm C** estimates the cost incurred when retrieval selects the wrong checklist.

The second contrast is the study's primary novel quantity. It is estimable only if the retriever is permitted to fail. Accordingly, retrieval error is treated as a property of the system under measurement and not as a defect to be engineered away. The corpus is held at full size for all queries, near-miss distractors are retained, and no per-case tuning, candidate filtering, module restriction, or post-hoc threshold selection is performed. A retriever achieving perfect accuracy on the case set would render Arm B equivalent to Arm C and eliminate the B → C contrast; perfect retrieval is therefore not a design objective.

R2. Frozen retrieval specification

The following parameters were fixed prior to generation and are held constant across all queries, cases, demographic variants, arms, and repeated runs.

Corpus. 32 clinical safety checklists, version v4 (final, locked). No corpus revisions are permitted after the lock date.

Indexed text. The `checklist_text` field only. The `checklist_id`, `topic`, and `role_in_study` fields are excluded from the index. Exclusion of `role_in_study` is required because that field contains study metadata (distractor designations), and its inclusion would leak experimental design information into the retrieval space.

Query text. The completed vignette text following demographic descriptor substitution — byte-identical to the text presented to the generating model. No diagnosis, gold checklist identifier, case identifier, case family, module label, or reviewer annotation is passed to the retriever.

Encoders. Two sentence-embedding models are evaluated:

Role	Model	Pinned revision
Primary	sentence-transformers/all-mpnet-base-v2	[hash]
Weak contrast	sentence-transformers/all-MiniLM-L6-v2	[hash]

Both are loaded through the `sentence-transformers` library, which applies mean pooling. Raw `transformers` loading with CLS pooling is not used, as the two produce different embeddings from identical weights. Model revisions are pinned by commit hash and reported.

The weak contrast is retained deliberately rather than as a baseline to be superseded. Retriever strength is not assumed monotonic across case difficulty (see R5).

Similarity and selection. Embeddings are L2-normalized and compared by cosine similarity. The rank-1 checklist is injected. Ranks 1–3, their similarity scores, and the rank-1-minus-rank-2 margin are logged for every query. Ties are broken deterministically by ascending `checklist_id`.

Determinism and caching. Retrieval is computed once per unique (case × demographic variant) query and cached to a manifest prior to generation. The manifest is the sole source of injected checklist assignments; retrieval is not recomputed during generation. Repeated runs therefore vary only in model sampling, not in retrieval. Agreement of `retrieval_selected_checklist_id` across all runs within a run group is verified as a data integrity check.

Cross-implementation verification. Two independent implementations were required to agree before the specification was locked. Agreement was established by embedding three fixed strings and comparing cosine similarities to four decimal places, then by confirming identical rank-1 selections and similarity scores across all queries. An earlier divergence between the two implementations was traced to differing index-string construction (inclusion versus exclusion of the `topic` prefix) and resolved by adopting the specification above. This verification procedure is reported as part of the methods.

R3. Retrieval outcome classification

Each query is classified against the gold checklist:

Label	Condition
Y	gold checklist at rank 1
PARTIAL	gold checklist in ranks 2–3
N	gold checklist outside ranks 1–3

Unit of analysis. Retrieval accuracy is reported per unique query, not per generated response. Because retrieval is deterministic under the frozen specification, repeated runs within a run group are not independent observations of retrieval performance, and counting them separately inflates the effective sample. For a design of k cases × d demographic variants, the retrieval denominator is $k \times d$.

R4. Margin stratification and demographic divergence

`rank1_minus_rank2_margin` is logged on every Arm B row. Analyses of Arm B outcomes are stratified into **near-tie** and **decided** selections, with the threshold prespecified at 0.01 cosine.

Rationale. In pilot data, every instance in which two demographic variants of the same case selected different checklists involved at least one variant with a margin below approximately 0.01, and no selection with both variants above that margin diverged. Where the margin between the top two candidates is small, the demographic descriptor is sufficient to change the selected checklist.

Interpretation. This is prespecified as **ranking instability under near-tied similarity**, not as directional demographic bias. In pilot data the direction of divergence was inconsistent across cases — some divergences moved toward a more clinically hazardous checklist and others toward a less hazardous one — and the number of affected cases does not support a directional claim.

Reporting commitment. Any Arm B outcome difference between demographic variants is reported conditional on retrieval stratum. Differences arising in the near-tie stratum are attributed to retrieval-layer instability and are distinguished from differences arising in the decided stratum, where the injected checklist is identical across variants and any difference is attributable to the generating model. Retrieval-layer divergence is disclosed rather than removed, since removing it (for example by fixing the injected checklist across variants) would eliminate the deployable-guardrail interpretation of Arm B.

R5. Prespecified indexing ablation

A secondary analysis compares two index-construction variants under otherwise identical settings:

- **Variant 1 (frozen specification):** `checklist_text` only.
- **Variant 2:** `topic` prefixed to `checklist_text`.

Hypothesis. Retrieval accuracy and the resulting Arm B safety outcomes are sensitive to index-string construction, a decision typically left undocumented in deployed retrieval-augmented systems.

Motivation. In pilot data the two variants produced different top-1 selections on multiple cases and different accuracy on the case set, with neither dominating: the prefix recovered some cases and lost others. This is reported as evidence that retrieval implementation choices — not only retriever model capacity — affect downstream safety-relevant output.

A related observation is prespecified for reporting: the weak contrast retriever outperformed the primary retriever on at least one edge case in pilot data. Retriever capacity is therefore not assumed to be monotonically related to retrieval correctness across case difficulty, and per-case retrieval outcomes are reported alongside aggregate accuracy.

R6. Exploratory: reasoning-token cost of conflicting guidance

For generating models that report reasoning token counts, `reasoning_tokens` is logged per response. A prespecified exploratory analysis compares reasoning token counts within case across three conditions: no checklist injected (Arm A), correct checklist injected, and incorrect checklist injected.

Motivation. In pilot data, injection of an incorrect checklist was associated with elevated reasoning token counts relative to the raw arm within the same case, while the final response text frequently contained no stated objection to the injected checklist. This raises the possibility that a model may detect a conflict between the case and the injected guidance without surfacing that conflict in its output — a state not observable to a reviewer scoring the final text alone.

Status. Exploratory and hypothesis-generating only. The effect was not uniform across cases in pilot data, the comparison is confounded with intrinsic case difficulty unless conducted within case, and the measure is unavailable for models that do not report reasoning tokens. No confirmatory claim is prespecified.

R7. Logged fields

The following are recorded for every Arm B response:

```
retrieval_selected_checklist_id   retrieval_gold_checklist_id
retrieval_top3_ids   retrieval_top3_scores
retrieval_selected_rank1_score   rank1_minus_rank2_margin
retrieval_correct   ADMIN_retrieval_version
ADMIN_guardrail_corpus_version
```

`ADMIN_retrieval_version` uniquely identifies the retriever configuration so that multiple Arm B conditions remain separable within a single data file. It is left blank for Arms A and C.

`ADMIN_temperature` records the requested sampling temperature and `ADMIN_temperature_honoured` records whether the endpoint applied it. Certain frontier model families do not expose a temperature parameter; requests to those endpoints succeed while the parameter is discarded upstream. Affected rows are flagged rather than silently recorded as temperature-controlled, and the limitation is disclosed.